CES-FL: Six Tables of Best Observed Results

Ding Tables1–3 and FairForge-2 split, validation and TEST

9 November2026

Approved retrospective best-observed summary. Each cell may come from a different model or seed. Best-observed rows are not one trained model, not unbiased estimates of future performance, and not evidence of an overall win. No new training, model inference or threshold search was performed.

1. Table1: Ding's FF++ intra-domain table, original published rows and columns, plus our cellwise best.
2. Table2: Ding's cross-domain Xception table, original published rows and columns, plus our cellwise best.
3. Table3: Ding's ResNet-50 table, original published rows and columns, plus our cellwise best.
4. Table4: Actual public FairForge-2 TRAIN, validation, TEST counts; reserved CAL disclosed separately.
5. Table5: FairForge-2 validation comparison; best recorded seed per metric and best-own across variants.
6. Table6: FairForge-2 TEST comparison with the same layout; unavailable evaluations remain blank.

Reading the values. All AUC/rates are multiplied by100. F_{FPR} is a sum of group deviations, not the overall FPR; lower disparity/DP/FPR/FNR is better, higher AUC/es-AUC/worst-group AUC is better. Groups require20 real and20 fake images in the stored evaluations. Different numbers of eligible groups make disparity magnitudes across axes non-comparable. Age/skin labels are estimates.

Our cell labels. Every selected cell names its source run/seed beneath the value. X/R denote Xception/ResNet-50. LA/LH are plain-head arms; A/B/C are archived encoder/invariance controls; R0/R1/R2/RN/RD are archived upgrade arms. Some low-disparity cells can come from a weak or almost constant classifier: the provenance index includes each contributing run's AUC/FPR/FNR. FairForge tags P/A/B/LA/LH/S/J denote CES-FL P/A/B, plain-head LA/LH, RF-S, RF-SJg; suffix is seed.

Protocol limits. Tables1–3 use the existing fixed-probability0.5 reports, never a chosen TEST threshold. FairForge tables use the existing validation-real FPR10 calibration, frozen for TEST. Training budgets, selection rules and support differ from published protocols. ResNet LA/LH FF++ has231 recorded fit/test identity/source-ID overlaps; independence is unverified. Full counterfactual CES-FL has not been established by these experiments.

Published reference: https://arxiv.org/html/2511.10150v5. The paper's "Ours" is labelled "Ding et al. (published)" to distinguish it from our rows. Published numeric values are unchanged. Full precision/source provenance accompanies the PDF in CSV/JSON.

Table 1. Ding's original FF++ intra-domain comparison

Published rows retain their original order, method identities, backbones and numeric values; paper-internal bibliography indices are omitted. Blue rows are our post-hoc best observed per metric; source tags identify different runs/seeds. No published row is a new reproduction.

Attributes	Methods	Backbone	Fairness Metrics(%) $F_{\text{FPR}}\downarrow$	$F_{\text{DP}}\downarrow$	*es*-AUC↑	Detection Metrics(%) AUC↑
Gender	Ori (published)	Xception	4.10	5.72	91.93	92.69
	DAG-FDD WACV'24	Xception	1.82	4.65	94.87	96.72
	DAW-FDD WACV'24	Xception	0.78	9.52	95.76	97.46
	PG-FDD CVPR'24	Xception	0.62	4.74	96.32	97.66
	Fairadapter ICASSP'25	ViT-L/14	4.16	12.21	67.85	71.50
	RSEF-FDD arxiv'25	Xception	0.57	8.55	94.91	97.09
	Ding et al. (published)	Xception	0.53	3.61	96.45	97.71
	CES-FL best observed, cellwise	Xception	**0.11** X-RN-17	**1.76** X-LA-17	**97.54** X-LA-7	**98.54** X-LA-7
Race	Ori (published)	Xception	19.76	4.74	82.85	92.69
	DAG-FDD WACV'24	Xception	5.48	9.20	93.43	96.72
	DAW-FDD WACV'24	Xception	5.43	14.69	94.15	97.46
	PG-FDD CVPR'24	Xception	11.13	4.78	94.52	97.66
	Fairadapter ICASSP'25	ViT-L/14	43.22	20.39	56.03	71.50
	RSEF-FDD arxiv'25	Xception	8.39	5.28	93.60	97.09
	Ding et al. (published)	Xception	9.29	4.35	94.86	97.71
	CES-FL best observed, cellwise	Xception	**6.03** X-LA-7	**2.66** X-R1-17	**97.57** X-LH-17	**98.54** X-LA-7
Intersection	Ori (published)	Xception	36.03	14.64	74.43	92.69
	DAG-FDD WACV'24	Xception	24.08	26.25	85.80	96.72
	DAW-FDD WACV'24	Xception	14.36	26.13	86.74	97.46
	PG-FDD CVPR'24	Xception	9.19	13.39	86.83	97.66
	Fairadapter ICASSP'25	ViT-L/14	86.91	42.44	45.98	71.50
	RSEF-FDD arxiv'25	Xception	23.64	21.74	85.77	97.09
	Ding et al. (published)	Xception	20.18	9.47	86.91	97.71
	CES-FL best observed, cellwise	Xception	**14.93** X-LA-7	**12.17** X-LA-17	**92.07** X-LH-17	**98.54** X-LA-7

AUC is repeated across attribute blocks in Table1. Source pools include all eligible archived CES-FL reports listed in the provenance index, plus newer LA/LH runs; fixed0.5 is held constant. A cellwise row can combine low disparity from one weak detector with high AUC from another. It is not a joint utility/fairness result.

Table 2. Ding's original cross-domain comparison

Published rows retain their original order, method identities, backbones and numeric values; paper-internal bibliography indices are omitted. Blue rows are our post-hoc best observed per metric; source tags identify different runs/seeds. No published row is a new reproduction.

			Fairness Metrics(%)									Detection Metrics(%)
			Gender			Race			Intersection			Overall
Datasets	Methods	Backbone	$F_{\text{FPR}}\downarrow$	$F_{\text{DP}}\downarrow$	*es*-AUC↑	$F_{\text{FPR}}\downarrow$	$F_{\text{DP}}\downarrow$	*es*-AUC↑	$F_{\text{FPR}}\downarrow$	$F_{\text{DP}}\downarrow$	*es*-AUC↑	AUC↑
DFDC	Ori (published)	Xception	8.67	6.70	54.56	19.41	7.99	47.02	53.42	17.34	36.96	56.13
	DAG-FDD WACV'24	Xception	5.29	6.68	57.41	9.26	12.51	45.91	45.30	12.92	36.42	59.13
	DAW-FDD WACV'24	Xception	3.60	3.67	55.88	22.36	9.72	47.77	46.80	12.69	36.63	56.90
	PG-FDD CVPR'24	Xception	2.35	2.57	57.71	10.10	11.49	50.53	32.16	17.71	37.42	59.64
	Fairadapter ICASSP'25	ViT-L/14	2.39	4.57	55.88	32.49	26.77	52.27	74.10	31.53	31.39	59.25
	RSEF-FDD arxiv'25	Xception	2.09	3.83	57.90	12.24	7.74	50.27	38.89	12.18	37.72	59.08
	Ding et al. (published)	Xception	1.76	3.67	57.94	17.93	7.58	52.33	37.37	11.92	39.03	61.86
	CES-FL best observed, cellwise	Xception	**0.00** X-LA-17	**1.42** X-LA-17	**60.69** X-LA-17	**6.41** X-LH-17	**4.51** X-LH-17	**53.49** X-LA-17	**15.02** X-LH-17	**6.63** X-LH-17	**42.87** X-C-17	**61.77** X-LA-17
Celeb-DF	Ori (published)	Xception	6.93	20.23	60.94	23.44	17.89	61.21	27.38	18.72	61.22	69.18
	DAG-FDD WACV'24	Xception	8.70	14.25	66.23	19.44	13.26	62.61	23.60	13.80	63.08	72.29
	DAW-FDD WACV'24	Xception	8.71	12.73	68.31	14.76	7.80	66.47	24.75	13.78	59.24	74.09
	PG-FDD CVPR'24	Xception	7.95	16.29	69.47	23.99	12.47	67.78	23.71	13.30	62.63	74.24
	Fairadapter ICASSP'25	ViT-L/14	8.59	26.84	56.84	46.14	36.35	58.34	47.03	35.69	54.77	64.94
	RSEF-FDD arxiv'25	Xception	1.94	35.63	68.92	17.40	27.48	67.80	24.17	27.73	62.56	74.36
	Ding et al. (published)	Xception	6.41	10.81	69.68	16.60	7.29	69.96	23.32	12.01	63.27	76.75
	CES-FL best observed, cellwise	Xception	**1.51** X-LA-7	**11.79** X-A-17	**64.48** X-R2-17	**15.03** X-A-17	**11.91** X-A-17	**59.66** X-LA-27	**20.62** X-A-17	**12.28** X-A-17	**58.18** X-A-17	**74.67** X-R2-17
DFD	Ori (published)	Xception	9.45	6.63	72.68	7.75	22.31	69.07	35.06	25.53	59.20	74.32
	DAG-FDD WACV'24	Xception	5.41	5.60	75.87	2.56	21.34	70.14	33.00	23.02	61.09	76.29
	DAW-FDD WACV'24	Xception	5.30	10.63	71.68	4.32	20.16	63.24	33.81	24.72	56.01	73.71
	PG-FDD CVPR'24	Xception	5.05	1.86	76.37	5.79	19.31	74.81	28.00	23.93	66.32	80.70
	Fairadapter ICASSP'25	ViT-L/14	6.32	7.55	63.28	11.29	12.28	56.28	29.56	33.11	48.63	68.12
	RSEF-FDD arxiv'25	Xception	18.50	6.11	75.97	19.45	19.40	72.65	35.34	10.97	67.24	80.54
	Ding et al. (published)	Xception	4.72	5.49	78.98	2.09	18.66	77.44	27.32	22.81	69.73	81.46
	CES-FL best observed, cellwise	Xception	**0.85** X-LH-17	**1.48** X-C-17	**83.79** X-LA-7	**1.96** X-LA-27	**0.82** X-RD-17	**82.25** X-LA-7	**15.54** X-LA-27	**4.93** X-RD-17	**76.52** X-RD-17	**84.23** X-LA-7

AUC is repeated across attribute blocks in Table1. Source pools include all eligible archived CES-FL reports listed in the provenance index, plus newer LA/LH runs; fixed0.5 is held constant. A cellwise row can combine low disparity from one weak detector with high AUC from another. It is not a joint utility/fairness result.

Table 3. Ding's original ResNet-50 comparison

Published rows retain their original order, method identities, backbones and numeric values; paper-internal bibliography indices are omitted. Blue rows are our post-hoc best observed per metric; source tags identify different runs/seeds. No published row is a new reproduction.

			Fairness Metrics(%)									Detection Metrics(%)
			Gender			Race			Intersection			Overall
Datasets	Methods	Backbone	$F_{\text{FPR}}\downarrow$	$F_{\text{DP}}\downarrow$	es-AUC↑	$F_{\text{FPR}}\downarrow$	$F_{\text{DP}}\downarrow$	es-AUC↑	$F_{\text{FPR}}\downarrow$	$F_{\text{DP}}\downarrow$	es-AUC↑	AUC↑
FF++	Ori (published)	ResNet-50	8.90	9.27	90.40	17.49	7.04	89.17	38.26	15.43	76.14	94.29
	DAG-FDD WACV'24	ResNet-50	1.92	6.02	91.09	6.38	3.98	91.39	32.58	11.06	80.64	94.74
	DAW-FDD WACV'24	ResNet-50	3.25	5.29	90.13	12.46	3.29	90.92	31.07	11.03	80.98	93.96
	PG-FDD CVPR'24	ResNet-50	3.05	3.06	93.57	10.89	2.83	91.53	22.10	9.69	81.58	94.46
	Ding et al. (published)	ResNet-50	3.94	5.14	94.46	18.67	2.43	92.69	34.30	8.61	82.87	95.53
	CES-FL best observed, cellwise	ResNet-50	**0.02** R-LH-17	**3.10** R-A-17	**93.72** R-LA-17	**14.86** R-LA-17	**2.67** R-LH-17	**92.77** R-LH-17	**30.43** R-LH-17	**8.93** R-LH-17	**82.79** R-C-17	**96.21** R-LA-17
DFDC	Ori (published)	ResNet-50	12.55	11.06	54.96	17.34	8.38	48.30	48.10	16.30	35.76	57.83
	DAG-FDD WACV'24	ResNet-50	6.57	4.87	57.63	15.96	2.95	50.31	44.10	11.25	33.41	60.29
	DAW-FDD WACV'24	ResNet-50	9.12	7.14	57.22	14.60	2.81	48.94	43.31	9.55	34.71	59.78
	PG-FDD CVPR'24	ResNet-50	5.78	4.65	58.34	12.68	5.23	49.33	31.97	8.54	34.85	61.83
	Ding et al. (published)	ResNet-50	3.97	3.61	56.96	11.94	5.15	50.99	32.14	10.27	37.55	62.54
	CES-FL best observed, cellwise	ResNet-50	**0.02** R-A-17	**0.31** R-A-17	**57.95** R-LH-17	**8.76** R-A-17	**1.60** R-LH-17	**51.18** R-LH-17	**30.42** R-A-17	**12.36** R-LA-17	**36.15** R-LA-17	**60.07** R-LH-17
Celeb-DF	Ori (published)	ResNet-50	12.26	22.12	61.11	18.55	22.43	59.82	24.84	23.18	56.09	71.65
	DAG-FDD WACV'24	ResNet-50	8.22	20.45	67.01	23.07	18.50	68.07	20.29	19.33	66.76	75.77
	DAW-FDD WACV'24	ResNet-50	9.03	20.85	68.33	18.60	17.65	69.14	23.84	18.55	68.74	76.51
	PG-FDD CVPR'24	ResNet-50	6.98	16.92	68.49	16.74	16.50	70.35	29.34	18.72	66.26	76.10
	Ding et al. (published)	ResNet-50	4.53	12.00	68.98	6.76	8.16	71.03	5.56	9.11	68.25	77.35
	CES-FL best observed, cellwise	ResNet-50	**4.26** R-A-17	**10.60** R-A-17	**63.16** R-B-17	**4.53** R-A-17	**6.48** R-A-17	**61.96** R-LH-17	**6.41** R-A-17	**7.11** R-A-17	**59.50** R-A-17	**73.81** R-LH-17
DFD	Ori (published)	ResNet-50	10.95	9.08	70.33	4.66	12.57	66.68	33.46	16.84	56.00	71.31
	DAG-FDD WACV'24	ResNet-50	3.62	4.19	73.74	3.86	13.54	70.33	25.22	14.84	57.19	73.97
	DAW-FDD WACV'24	ResNet-50	4.81	3.90	72.12	7.36	3.21	70.57	33.45	14.30	54.52	73.58
	PG-FDD CVPR'24	ResNet-50	3.57	1.82	76.14	2.62	7.81	72.16	23.15	11.01	62.65	77.47
	Ding et al. (published)	ResNet-50	3.12	1.76	77.68	6.51	3.03	70.93	20.99	4.85	66.19	78.63
	CES-FL best observed, cellwise	ResNet-50	**2.10** R-A-17	**1.31** R-LA-17	**72.32** R-LA-17	**6.63** R-LH-17	**0.95** R-LH-17	**71.72** R-C-17	**23.95** R-A-17	**6.31** R-LH-17	**56.51** R-LA-17	**74.02** R-C-17

AUC is repeated across attribute blocks in Table1. Source pools include all eligible archived CES-FL reports listed in the provenance index, plus newer LA/LH runs; fixed0.5 is held constant. A cellwise row can combine low disparity from one weak detector with high AUC from another. It is not a joint utility/fairness result.

Table 4. FairForge-2 public split used in these experiments

Split	Real images	Fake images	Total images
TRAIN	16,777	201,489	218,266
Validation (SELECT)	2,438	28,669	31,107
TEST	5,441	82,123	87,564
Reserved calibration (CAL)	2,509	29,746	32,255
TRAIN + validation + TEST	24,656	312,281	336,937

Recorded gender×race composition (real / fake images)

Group	TRAIN	Validation (SELECT)	TEST	Reserved CAL
Male-Asian	2,219 / 26,500	322 / 3,833	722 / 11,573	324 / 3,950
Male-White	1,809 / 23,236	256 / 3,178	612 / 10,654	243 / 2,916
Male-Black	1,795 / 23,407	238 / 3,072	585 / 10,261	248 / 3,148
Male-Others	4,171 / 46,860	621 / 6,973	1,340 / 15,223	705 / 7,934
Female-Asian	1,333 / 17,079	202 / 2,493	449 / 8,711	194 / 2,479
Female-White	1,774 / 21,658	257 / 2,966	567 / 9,348	232 / 2,817
Female-Black	1,524 / 18,463	194 / 2,272	475 / 8,392	205 / 2,469
Female-Others	2,152 / 24,286	348 / 3,882	691 / 7,961	358 / 4,033

Counts describe the public cell-valid subset and manifests used by the reported models, not every historical/private FairForge-2 image. TRAIN/validation/TEST counts were independently checked against their actual manifests. Reserved CAL counts come from the saved public release composition; CAL was not substituted into the completed selection/calibration protocol. SELECT served both validation selection and threshold calibration. Counts are images, not independent people.

Table 5. FairForge-2 validation: best recorded values

Every method row uses its best available recorded seed independently per metric. Singleton rows are seed17. Tagged values in multiseed and best-own rows identify the contributing seed/run. No threshold, checkpoint or model was changed.

(A) Detection and Ding-style demographic metrics

		Detection / gender×race WG				Gender			Race			Gender×race		
Model	Seed pool	AUC↑	FPR↓	FNR↓	WG-AUC↑	F_{FPR} ↓	F_{DP} ↓	*es*-AUC↑	F_{FPR} ↓	F_{DP} ↓	*es*-AUC↑	F_{FPR} ↓	F_{DP} ↓	*es*-AUC↑
BCE	17	99.80	9.97	0.30	99.60	3.43	0.11	99.74	19.54	1.52	99.22	40.52	1.62	98.61
DAG-FDD	17	99.81	9.97	0.22	99.37	4.28	0.18	99.66	18.32	1.51	99.26	37.89	1.68	98.50
DAW-FDD	17	99.84	9.97	0.24	99.49	3.43	0.09	99.80	12.87	1.29	99.38	30.70	1.56	98.85
Ding corrected GDA	17	99.06	9.97	1.07	98.06	0.72	0.24	98.98	4.09	1.38	97.21	12.44	1.72	95.31
CES-FL P	17	99.81	9.97	0.38	99.58	3.60	0.24	99.80	12.16	1.06	99.44	28.39	1.53	99.08
CES-FL A	17	99.81	9.97	0.33	99.60	3.26	0.14	99.79	12.80	1.12	99.43	29.10	1.43	99.05
CES-FL B	17	99.81	9.97	0.37	99.59	3.43	0.19	99.80	13.17	1.11	99.42	29.55	1.32	99.04
L_A (plain-head BCE)	17	99.81	9.97	0.28	99.52	4.11	0.20	99.77	17.54	1.33	99.22	37.37	1.58	98.66
L_H-PS (ours)	17	99.75	9.97	0.38	99.41	4.28	0.27	99.67	17.55	1.30	99.02	36.23	1.50	98.27
RF-S (ours)	7/17/27	**99.81** S-27	**9.97** S-17	**0.28** S-7	**99.52** S-27	**4.11** S-17	**0.20** S-7	**99.78** S-27	**16.02** S-17	**1.24** S-27	**99.24** S-27	**34.12** S-17	**1.47** S-7	**98.68** S-27
RF-SJg (ours)	7/17/27	**99.76** J-7	**9.97** J-17	**0.45** J-7	**99.51** J-7	**3.26** J-17	**0.21** J-27	**99.75** J-7	**13.05** J-17	**0.83** J-17	**99.17** J-7	**27.53** J-17	**1.16** J-27	**98.59** J-7
Best own observed, cellwise	mixed	**99.81** P-17	**9.97** A-17	**0.28** S-7	**99.60** A-17	**3.26** A-17	**0.14** A-17	**99.80** P-17	**12.16** P-17	**0.83** J-17	**99.44** P-17	**27.53** J-17	**1.16** J-27	**99.08** P-17

(B) Age, skin and additional intersections

		Age (estimated)			Skin (estimated)			Age×skin			Sex×age×race		
Model	Seed pool	F_{FPR} ↓	*es*-AUC↑	WG-AUC↑	F_{FPR} ↓	*es*-AUC↑	WG-AUC↑	F_{FPR} ↓	*es*-AUC↑	WG-AUC↑	F_{FPR} ↓	*es*-AUC↑	WG-AUC↑
BCE	17	8.74	99.61	99.69	14.57	99.50	99.66	78.64	97.92	99.39	27.64	98.96	99.54
DAG-FDD	17	10.45	99.58	99.72	12.78	99.40	99.69	74.22	97.87	99.58	33.73	99.04	99.58
DAW-FDD	17	10.45	99.59	99.72	6.80	99.62	99.73	49.75	98.43	99.31	34.15	99.30	99.66
Ding corrected GDA	17	1.58	98.41	98.69	7.72	97.46	98.29	47.54	91.82	97.41	14.42	95.98	98.24
CES-FL P	17	9.23	99.63	99.70	11.68	99.62	99.70	75.00	98.59	99.28	25.98	99.38	99.65
CES-FL A	17	9.02	99.57	99.67	10.62	99.61	99.71	69.21	98.42	99.30	29.57	99.22	99.60
CES-FL B	17	7.42	99.60	99.68	11.56	99.62	99.69	70.98	98.48	99.28	27.30	99.32	99.62
L_A (plain-head BCE)	17	9.58	99.69	99.73	14.23	99.53	99.67	80.64	98.16	99.54	24.89	99.01	99.59
L_H-PS (ours)	17	11.90	99.40	99.60	13.35	99.30	99.61	80.63	97.11	99.15	35.96	98.90	99.46
RF-S (ours)	7/17/27	**9.17** S-7	**99.70** S-27	**99.74** S-27	**13.04** S-27	**99.54** S-27	**99.68** S-27	**73.60** S-17	**98.14** S-17	**99.56** S-17	**23.28** S-17	**98.97** S-17	**99.59** S-17
RF-SJg (ours)	7/17/27	**7.13** J-27	**99.63** J-17	**99.70** J-7	**8.15** J-27	**99.44** J-7	**99.60** J-7	**64.23** J-17	**97.87** J-17	**99.43** J-17	**28.97** J-17	**98.87** J-17	**99.50** J-17
Best own observed, cellwise	mixed	**7.13** J-27	**99.70** S-27	**99.74** S-27	**8.15** J-27	**99.62** P-17	**99.71** A-17	**64.23** J-17	**98.59** P-17	**99.56** S-17	**23.28** S-17	**99.38** P-17	**99.65** P-17

Stored threshold rule: validation-real empirical FPR10 for each selected checkpoint. Best-own pool: CES-FL P/A/B, LA/LH-PS, RF-S and RF-SJg where actually evaluated. GDA is the corrected local Ding-style adaptation, not a published-paper score. Budgets and selection policies differ across families. RF seeds share one pretrained encoder. In validation, RF age×skin and coarse eight-group metrics are available only for seed17; no missing seed-axis results were fabricated. Eight-group axis uses Youth<30/Adult and White/NonWhite. Full counterfactual CES-FL, PG-FDD and verified F3Net/CORE TEST results are unavailable. No uncertainty is attached to these post-hoc cellwise extrema; they are not means or confidence bounds.

Table 6. FairForge-2 TEST: best recorded values

Every method row uses its best available recorded seed independently per metric. Singleton rows are seed17. Tagged values in multiseed and best-own rows identify the contributing seed/run. No threshold, checkpoint or model was changed.

(A) Detection and Ding-style demographic metrics

		Detection / gender×race WG				Gender			Race			Gender×race		
Model	Seed pool	AUC↑	FPR↓	FNR↓	WG-AUC↑	F_{FPR} ↓	F_{DP} ↓	*es*-AUC↑	F_{FPR} ↓	F_{DP} ↓	*es*-AUC↑	F_{FPR} ↓	F_{DP} ↓	*es*-AUC↑
BCE	–	Not evaluated on this split in the saved reports												
DAG-FDD	–	Not evaluated on this split in the saved reports												
DAW-FDD	17	98.87	14.41	2.29	97.22	2.72	0.05	98.32	22.24	3.33	95.41	45.35	4.39	91.98
Ding corrected GDA	17	93.08	11.38	7.80	89.03	0.36	1.86	91.19	3.29	8.52	82.19	9.47	9.44	73.24
CES-FL P	17	98.78	12.39	2.92	96.85	2.50	0.25	98.16	20.42	2.75	95.27	42.23	4.02	91.79
CES-FL A	–	Not evaluated on this split in the saved reports												
CES-FL B	–	Not evaluated on this split in the saved reports												
L_A (plain-head BCE)	17	99.00	15.14	1.44	97.33	4.02	0.41	98.46	31.15	2.81	95.46	63.67	2.96	91.86
L_H-PS (ours)	–	Not evaluated on this split in the saved reports												
RF-S (ours)	7/17/27	**99.00** S-7	**14.80** S-17	**1.46** S-7	**97.31** S-7	**3.93** S-7	**0.32** S-27	**98.45** S-7	**28.99** S-27	**2.79** S-7	**95.44** S-7	**59.73** S-27	**3.00** S-7	**91.83** S-7
RF-SJg (ours)	7/17/27	**98.67** J-7	**13.40** J-27	**2.59** J-7	**96.56** J-7	**4.22** J-7	**0.06** J-17	**98.07** J-7	**20.82** J-27	**3.90** J-7	**94.63** J-7	**43.72** J-27	**4.43** J-7	**90.62** J-7
Best own observed, cellwise	mixed	**99.00** LA-17	**12.39** P-17	**1.44** LA-17	**97.33** LA-17	**2.50** P-17	**0.06** J-17	**98.46** LA-17	**20.42** P-17	**2.75** P-17	**95.46** LA-17	**42.23** P-17	**2.96** LA-17	**91.86** LA-17

(B) Age, skin and additional intersections

		Age (estimated)			Skin (estimated)			Age×skin			Sex×age×race		
Model	Seed pool	F_{FPR} ↓	*es*-AUC↑	WG-AUC↑	F_{FPR} ↓	*es*-AUC↑	WG-AUC↑	F_{FPR} ↓	*es*-AUC↑	WG-AUC↑	F_{FPR} ↓	*es*-AUC↑	WG-AUC↑
BCE	–	Not evaluated on this split in the saved reports											
DAG-FDD	–	Not evaluated on this split in the saved reports											
DAW-FDD	17	12.53	98.43	98.69	16.86	96.09	97.57	91.00	88.55	97.34	62.90	96.28	97.76
Ding corrected GDA	17	4.37	90.87	92.39	3.44	81.59	88.31	31.97	60.08	87.60	12.86	83.92	90.75
CES-FL P	17	8.17	98.34	98.53	16.55	95.94	97.50	80.49	87.94	97.25	33.12	95.94	97.80
CES-FL A	–	Not evaluated on this split in the saved reports											
CES-FL B	–	Not evaluated on this split in the saved reports											
L_A (plain-head BCE)	17	13.25	98.48	98.82	21.80	96.24	97.70	102.75	88.59	96.73	57.26	96.53	98.10
L_H-PS (ours)	–	Not evaluated on this split in the saved reports											
RF-S (ours)	7/17/27	**12.21** S-27	**98.48** S-7	**98.83** S-7	**20.01** S-27	**96.23** S-7	**97.71** S-7	**96.17** S-27	**88.59** S-7	**96.76** S-7	**52.14** S-27	**96.58** S-27	**98.09** S-7
RF-SJg (ours)	7/17/27	**12.52** J-17	**98.26** J-7	**98.53** J-7	**14.56** J-27	**95.65** J-7	**97.39** J-7	**84.29** J-7	**87.50** J-7	**96.53** J-7	**50.53** J-17	**96.74** J-27	**97.83** J-7
Best own observed, cellwise	mixed	**8.17** P-17	**98.48** S-7	**98.83** S-7	**14.56** J-27	**96.24** LA-17	**97.71** S-7	**80.49** P-17	**88.59** LA-17	**97.25** P-17	**33.12** P-17	**96.74** J-27	**98.10** LA-17

Stored threshold rule: each checkpoint's frozen validation-real FPR10 threshold; achieved TEST FPR is shown, not guaranteed10%. Best-own pool: CES-FL P/A/B, LA/LH-PS, RF-S and RF-SJg where actually evaluated. GDA is the corrected local Ding-style adaptation, not a published-paper score. Budgets and selection policies differ across families. RF seeds share one pretrained encoder. In validation, RF age×skin and coarse eight-group metrics are available only for seed17; no missing seed-axis results were fabricated. Eight-group axis uses Youth<30/Adult and White/NonWhite. Full counterfactual CES-FL, PG-FDD and verified F3Net/CORE TEST results are unavailable. No uncertainty is attached to these post-hoc cellwise extrema; they are not means or confidence bounds.

Provenance index and interpretation

The following index is supporting metadata, not a seventh comparison table. It exposes the detection performance of every benchmark candidate so near-zero disparity from a low-utility detector is visible. The complete per-cell ledger, full-precision metrics and hashes are saved next to this PDF.

Source tag	Dataset	AUC(%)	FPR(%)	FNR(%)	Images
R-A-17	Celeb-DF	66.44	80.80	5.57	28,458
R-A-17	DFD	50.20	56.82	42.01	9,385
R-A-17	DFDC	56.41	66.17	26.78	22,857
R-A-17	FF++	71.13	45.92	25.04	25,401
R-B-17	Celeb-DF	72.80	69.88	8.93	28,458
R-B-17	DFD	73.91	51.29	18.88	9,385
R-B-17	DFDC	58.07	55.29	35.11	22,857
R-B-17	FF++	95.98	18.84	5.46	25,401
R-C-17	Celeb-DF	72.81	68.04	9.67	28,458
R-C-17	DFD	74.02	48.90	19.74	9,385
R-C-17	DFDC	58.37	51.75	37.95	22,857
R-C-17	FF++	95.89	17.58	6.02	25,401
R-LA-17	Celeb-DF	73.50	62.30	13.58	28,458
R-LA-17	DFD	73.65	52.46	18.05	9,385
R-LA-17	DFDC	59.54	35.57	51.23	22,857
R-LA-17	FF++	96.21	15.38	6.58	25,401
R-LH-17	Celeb-DF	73.81	65.83	11.42	28,458
R-LH-17	DFD	73.47	52.55	17.62	9,385
R-LH-17	DFDC	60.07	36.79	48.53	22,857
R-LH-17	FF++	96.05	16.15	6.51	25,401
X-A-17	Celeb-DF	64.28	78.26	7.52	28,458
X-A-17	DFD	59.31	64.10	23.18	9,385
X-A-17	DFDC	50.63	72.10	25.07	22,857
X-A-17	FF++	67.48	55.31	21.01	25,401
X-B-17	Celeb-DF	74.33	61.05	10.89	28,458
X-B-17	DFD	82.36	32.73	18.40	9,385
X-B-17	DFDC	59.47	21.29	65.63	22,857
X-B-17	FF++	98.35	8.57	5.39	25,401
X-C-17	Celeb-DF	73.70	54.00	14.39	28,458
X-C-17	DFD	81.24	25.97	22.25	9,385
X-C-17	DFDC	59.67	15.86	73.13	22,857
X-C-17	FF++	98.22	7.07	6.35	25,401
X-LA-17	Celeb-DF	70.75	61.09	15.70	28,458
X-LA-17	DFD	81.16	46.38	14.43	9,385
X-LA-17	DFDC	61.77	30.49	55.22	22,857
X-LA-17	FF++	98.47	9.37	3.88	25,401
X-LA-27	Celeb-DF	70.45	52.13	21.46	28,458
X-LA-27	DFD	81.20	54.85	10.34	9,385
X-LA-27	DFDC	61.50	30.03	55.01	22,857
X-LA-27	FF++	98.36	6.65	5.62	25,401
X-LA-7	Celeb-DF	71.27	58.80	15.52	28,458
X-LA-7	DFD	84.23	46.47	10.95	9,385
X-LA-7	DFDC	59.97	33.09	51.63	22,857
X-LA-7	FF++	98.54	8.40	4.20	25,401
X-LH-17	Celeb-DF	71.40	41.73	27.40	28,458
X-LH-17	DFD	80.88	31.86	22.47	9,385
X-LH-17	DFDC	60.46	18.04	70.64	22,857
X-LH-17	FF++	98.49	4.32	7.07	25,401
X-R0-17	Celeb-DF	74.33	61.05	10.89	28,458
X-R0-17	DFD	82.36	32.73	18.40	9,385
X-R0-17	DFDC	59.47	21.29	65.63	22,857
X-R0-17	FF++	98.35	8.57	5.39	25,401

Source tag	Dataset	AUC(%)	FPR(%)	FNR(%)	Images
X-R1-17	Celeb-DF	74.21	59.86	11.55	28,458
X-R1-17	DFD	81.81	35.22	18.16	9,385
X-R1-17	DFDC	58.97	23.53	62.00	22,857
X-R1-17	FF++	98.40	8.35	5.31	25,401
X-R2-17	Celeb-DF	74.67	67.13	8.49	28,458
X-R2-17	DFD	82.12	42.17	15.40	9,385
X-R2-17	DFDC	59.52	26.84	58.76	22,857
X-R2-17	FF++	98.40	9.32	4.86	25,401
X-RD-17	Celeb-DF	72.80	57.01	14.11	28,458
X-RD-17	DFD	82.28	37.06	17.27	9,385
X-RD-17	DFDC	59.86	22.36	65.48	22,857
X-RD-17	FF++	98.20	8.84	5.20	25,401
X-RN-17	Celeb-DF	71.01	59.18	11.91	28,458
X-RN-17	DFD	81.53	32.44	21.15	9,385
X-RN-17	DFDC	59.67	24.58	63.58	22,857
X-RN-17	FF++	97.85	9.06	5.25	25,401

A/B/C: original frozen ImageNet / detection-pretrained / invariance controls. R0/R1/R2/RN/RD: archived frozen BCE / partial BCE / group-DRO / no-change / age pilots, seed17. R0 repeats the archived B results and is not independent replication. LA/LH: newer plain-head matched arms; X-LA has seeds7/17/27, X-LH and R-LA/LH seed17. Cells using an almost constant classifier do not demonstrate useful fairness. DFDC retains1,476 excess duplicate rows in22,857 rows (21,381 unique images). Published and our scores are descriptive comparisons with unresolved protocol differences.